Fiscal Year:	FY 2026	**Task Last Updated:**	FY 12/19/2025
PI Name:	Christenson, Lane Ph.D.		
Project Title:	Female Reproductive Health: Space Flight Induced Ovarian and Estrogen Signaling Dysfunction, Adaptation, and Recovery--80NSSC24M0072 ; NNX15AB48G		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology (2) Developmental Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	66160-0001	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2014 Space Biology Flight NNH14ZTT001N
Start Date:	01/25/2018	**End Date:**	01/31/2027
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA ARC
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 01/31/2027 per F. Hernandez/ARC (Ed., 3/11/26). NOTE: End date changed to 1/31/2026 per F. Hernandez/NASA-ARC (Ed., 2/12/25). NOTE: New grant number (80NSSC24M0072) per NSSC information (Ed., 2/3/25). NOTE: End date changed to 1/31/2024 per NSSC information (Ed., 11/6/2023) NOTE: End date changed to 1/31/2023 per NSSC information (Ed., 1/24/2022) NOTE: End date changed to 1/31/2022 per NSSC information (Ed., 1/22/2021) NOTE: End date changed to 1/31/2021 per F. Hernandez/ARC (Ed., 2/3/2020) NOTE: End date changed to 1/31/2020 per NSSC information (Ed., 3/12/19)		

Key Personnel Changes/Previous PI:	We have added Dr. Stephanie Puukila from the NASA Ames Research Center as a CoI in June of 2023. Dr. Puukila was added due to expertise in behavioral analyses and due to our addition of a study looking at the behavior of the offspring born to the Flight (FLT) females. This work is a natural extension of Aim 3 Specific Aim, where we are evaluating fertility of the FLT mice, to now include whether those mice born to FLT mice are normal with respect to behavior.
COI Name (Institution):	Ronca, April Ph.D. (NASA Ames Research Center) Alwood, Josh Ph.D. (NASA Ames Research Center) Puukila, Stephanie A (NASA Ames Research Center)
Grant/Contract No.:	80NSSC24M0072 ; NNX15AB48G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	Continuation of "Female Reproductive Health: Space Flight Induced Ovarian and Estrogen Signaling Dysfunction, Adaptation, and Recovery" with the grant number NNX15AB48G and Principal Investigator (PI) Lane K. Christenson, Ph.D. (Previous PI was Joseph Tash, Ph.D.). In view of NASA's plans to conduct multigenerational studies on rodents in long-term spaceflight, these observations raise important fundamental questions on maintenance of female reproductive health that need to be addressed prior to conducting complex multigenerational studies in mammals under spaceflight conditions. As a result of our previous findings, specific important knowledge gaps now exist regarding the impact of spaceflight on female reproductive health. The central hypothesis that we will test is that spaceflight elicits direct and/or indirect effects at the ovarian level to cause infertility in adult female mice. This grant application combines the expertise of Dr. Christenson in ovarian physiology, and Dr. Ronca in pregnancy and early development, to address this central hypothesis. Additionally, these studies will address an important secondary hypothesis that the loss in uterine estrogen receptor expression will also occur in bone and this may exacerbate the observed bone mass loss during spaceflight (Dr. Alwood as collaborator). To address these hypotheses the following specific aims are proposed: Aim 1. Determine whether short-term (~10 d) spaceflight-disrupted ovarian function is due to a direct effect at the ovarian level. The working hypothesis, based on our preliminary data, is that disrupted ovarian function is due to a failure of the periovulatory follicle to ovulate and form a functional corpus luteum. To test this hypothesis, mice will undergo a timed follicular stimulation protocol followed by evaluation of established temporally expressed ovarian markers, ovulation, and luteal development. This study will for the first time determine if space induced loss of ovarian function is due to an intrinsic ovarian defect or is a problem with hypothalamic/pituitary support. Aim 2. Determine if adult female mice recover ovarian function during extended (~60 d) flight time. Currently, it is unknown whether females will recover fertility following long-term (i.e., ~60 d) acclimation to spaceflight, or will they remain infertile or exhibit more pronounced defects in ovarian/uterine function. To address this question, mice will undergo a similar follicular stimulation and ovulatory protocol in flight but after ~60 d, as described in Aim 1, and fertility and tissue biomarkers will be evaluated. Aim 3. Determine if adult female mice exposed to spaceflight are fertile upon return to Earth. The working hypothesis for this aim is that upon return to Earth, fertility will return. To test this hypothesis, flight mice will be exposed to male mice during a recovery period upon return to Earth, and regain of mating reproductive behavior, estrous cyclicity, early embryo implantation, and health and development of F1 offspring will be evaluated. These studies will provide the first ever examination of long-term space exposure on the female reproductive behavior/physiology, and germ-line stability. In all 3 specific aims, collection of uterine and bone (femur) tissues will allow us to address our secondary hypothesis by determining long-term spaceflight effects on estrogen receptor expression. Ultimately, findings of the proposed studies will provide critical foundational data to support multigenerational mammalian studies in space, and importantly, aid in our understanding of the impact spaceflight has on female astronaut health.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Fertility is a general measure of fitness of a species on Earth. Disruptions to fertility have both short- and long-term effects on the health on the individual or the ability of a species to maintain critical numbers, respectively. Earth species have adapted to changes in environment over millennia; these adaptations allow species to survive and flourish on Earth. The reproductive systems of both female and male mammals are well tuned to sense environmental changes. These environmental responses can then either increase or decrease fertility appropriately. The effects of weightlessness on female ovarian function we currently are studying are critical as we consider longer-term flights and the overall health of these individuals upon return to Earth. The fact that ovarian estrogen plays a key metabolic role in almost all tissues of the body also highlights the importance of understanding how female fertility is influenced by exposure to space. The Earth benefits of this research may yield new, yet undiscovered mechanisms controlling ovarian function, steroidogenesis, and female fertility. Our research has led us to adapt an approach to monitor estrous cyclicity in rodents; we believe this methodology has the potential to impact all female based NASA flight research projects due to our ability to accurately determine the stage of the estrous cycle and thus predict ovarian estrogen levels.
Task Progress:	Rodent Research-20 launched on November 9, 2023 and half of the females returned from the International Space Station (ISS) on December 23, 2023. These female mice were mated to ground-based males and reproductive fitness/fertility (litter size and ovarian and uterine histology) were examined and litter size was slightly impacted. Subsequent analysis of the pups, however, indicated an increase in weight gain for female offspring from spaceflight females compared to ground females. Behavior and tissue morphology, hormone production were evaluated in F1 offspring, and both male and female offspring exhibited growth and behavorial differences that were retained into adulthood. Indeed, female offspring from spaceflight mice exhibited premature ovarian insufficiency (POI), which is characterized by early loss of fertility due to reduced numbers of oocytes within the ovary. Moreover, when these F1 offspring were allowed to mate, the resultant F2 offspring (grandpups of the spaceflight females), again had reduced fertility longevity – and when these animals were provided an additional stress in the form of a high fat/high sucrose, Western-style diet – male pups, whose grandmother flew to space, exhibited increased obesity compared to pups that were ground-based. The ability of females to ovulate in space was also tested and no differences were observed in

	spaceflight and ground females, interestingly; though the uterus of the spaceflight mice exhibited a marked reduction in size, which we are currently testing to see if this is due to reduced estrogen hormone production or due to a reduced ability of the tissue to respond to the hormone. Ongoing studies are looking at whether differences in the ovary and uterus can be identified using spatial analysis of gene expression. The current finding indicates a generational impact of spaceflight can be passed down to subsequent generations.
Bibliography Type:	Description: (Last Updated: 01/03/2026)
Abstracts for Journals and Proceedings	Ahmed A, Figueiredo D, Quilici D, Woolsey R, Alwood JS, Ronca AE, Christenson LK, Tahimic C. "Role of Redox Signaling n the organismal response to spaceflight: insights from the heart proteome. " 2025 ASGSR Meeting, Phoenix, AZ December 3-6, 2025 Poster Presentation Abstracts 2025 ASGSR Meeting, Phoenix, AZ December 3-6, 2025 , Dec-2025
Abstracts for Journals and Proceedings	Espinosa E, Ramierz YC, Dvorochkin T, Puukila S, Tabares-Ruiz S, Christenson LK, Ronca AE, Alwood JS. "Activity and behavior of female mice during 24 hour light and dark cycle of RR-20 long-duration spaceflight mission. " 2025 ASGSR Meeting, Phoenix AZ, December 3-6, 2025. Oral Presentation Abstract. 2025 ASGSR Meeting, Phoenix AZ, December 3-6, 2025. , Dec-2025
Abstracts for Journals and Proceedings	Stroud M, Meruelo MT, Siceloff S, Mitchell C, Juran C, Christenson LK, Alwood JS, Allen N, Blaber E, Paul A. "Immune Influence on Reproductive Health Following Spaceflight" 2025 ASGSR Meeting, Phoenix AZ, December 3-6, 2025. Oral Presentation Abracts. 2025 ASGSR Meeting, Phoenix AZ, December 3-6, 2025. , Dec-2025
Articles in Other Journals or Periodicals	Hong X, Prom J, Ponte M, Dalouchi F, Hockey B, Braun J, Nies P, Stoltz A, Choi S, Ronca A, Alwood J, Fajardo V, Morris E, Puukila S, Christenson LK. "Impact of Female Space Travel on Progeny Health" Research Square Preprint Posted May 29th, 2025 https://doi.org/10.21203/rs.3.rs-6458038/v1 , May-2025
Conference Materials (Downloadable)	Dalouchi F, Puukila S, Ronca AE, Hong X, Morris EM, Prom J, Ponte M, Hong X, Alwood JS, Christenson LK. "Transgenerational Effects of Spaceflight on Mammalian Offspring: Behavioral and Metabolic Outcomes." Society for Study of Reproduction Annual Meeting, Washington DC, July 27- August 2, 2025. , Jul-2025
Conference Materials (Downloadable)	Aguilera-Gallegos T, Higgins L, Chakrravarthi VP, Christenson LK. "Impact of Galactic Cosmic Rays on Ovarian Follicle Development and its Application in Cancer Biology" 2025 ACE Sypmosia, Kansas City, KS, July 25, 2025. , Jul-2025
Conference Materials (Downloadable)	Sinha M, Christenson LK. "Trangenerational Impacts of Spaceflight on Aging" Gerapalooza 2025, Landon Center on Aging, Kansas City, KS October 16, 2025 , Oct-2025
Conference Materials (Downloadable)	Siceloff SI, Madrigal AC, Harmon C, Poague H, Mazumdar R, Zhang R, Auilera-Gallegos T, Sinha M, Christenson LK. "Uterine Morphology is Altered During Extended Spaceflight on the International Space Station" 2025 ASGSR Meeting, Phoenix, AZ, December 3-6, 2025 , Dec-2025
Conference Materials (Downloadable)	Sinha M, Dalouchi F, Siceloff S, Hong X, Puukila S, Ronca AE, Alwood JS, Christenson LK. "Long-Duration Spaceflight Impairs Ovarian Function and Reproductive Outcomes in Successive Generations" 2025 ASGSR Meeting, Phoenix, AZ December 3-6, 2025. Oral Presentation , Dec-2025